# Huawei Access Node ELEC-ACC-57 Hardware/Environmental Fault Recovery Procedure

**Node:** ELEC-ACC-57 | **Vendor:** Huawei | **Severity:** Major | **Alarm Code:** ALARM_1006

## Description

The failure domain of Hardware/Environmental Fault, as denoted by alarm code ALARM_1006 on the Huawei access node ELEC-ACC-57 located at Electronic City Hub, typically occurs due to intrinsic ASIC memory parity errors, line card crashes, or power supply redundancy loss. This is often a consequence of the complex interplay between the node's control and data planes, where the ASIC (Application-Specific Integrated Circuit) plays a pivotal role in managing packet forwarding and control protocols. The parity errors in ASIC memory can lead to system instability, while line card crashes can disrupt network services. Moreover, the loss of power supply redundancy can cause node failures, leading to significant network downtime. Understanding the underlying mechanics of the Huawei equipment and the specific role of the access node in the network architecture is crucial for diagnosing and resolving such issues. The access node, being responsible for aggregating traffic from various sources and forwarding it to the core network, must maintain high availability and reliability, making prompt and accurate troubleshooting essential.

## Pre-requisite Checks

- Dispatch remote hands to Electronic City Hub for physical inspection and intervention if necessary.

- Verify the status of redundant links to ensure no single point of failure exists.

- Check the environmental conditions (temperature, humidity) at the site to ensure they are within the recommended range for the Huawei equipment.

## Potential Root Causes

| Cause | Probability |
|---|---|
| ASIC Memory Parity Errors | High |
| Line Card Crashes | Medium |
| Power Supply Redundancy Loss | Low |

## Diagnostic Commands & Expected Outputs

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Command: display diagnostic-information, Expected Output: Detailed diagnostic information about the node's hardware and software status., Description: This command provides a comprehensive overview of the node's current state, including any errors or warnings that could indicate the root cause of the failure.

Command: display cpu-usage, Expected Output: Current CPU usage statistics., Description: High CPU usage could indicate software-related issues or excessive control plane traffic, which
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might contribute to the observed failures.

Command: display environment, Expected Output: Environmental parameters such as temperature, voltage, and fan status., Description: Abnormal environmental conditions can lead to hardware failures; this command helps in identifying any such issues.

## Resolution Steps

Step 1: Step: Replace Faulty Line Card, Command: replace board , Description: If the diagnostic commands indicate a faulty line card, replacing it with a spare can resolve the issue. Ensure the new line card is compatible with the existing configuration and is properly seated and secured.

Step 2: Step: Reset ASIC, Command: reset asic , Description: In cases where ASIC memory parity errors are identified, resetting the ASIC can sometimes clear the errors and restore normal operation. However, this should be done with caution as it may lead to temporary service disruption.

Step 3: Step: Restore Power Supply Redundancy, Command: power supply enable, Description: If the power supply redundancy is lost, enabling the standby power supply can restore redundancy and prevent future node failures due to power issues.

## Post-Check Verification

Command: display alarm, Expected Output: Current alarm status, ideally showing the clearance of ALARM_1006., Description: After executing the resolution steps, verify that the alarm indicating the hardware/environmental fault has been cleared, confirming the resolution of the issue.

Command: display diagnostic-information, Expected Output: Updated diagnostic information showing no errors or warnings related to the previously observed failure., Description: Run the diagnostic commands again to ensure that the node's hardware and software are in a healthy state and that the failure has been successfully resolved.

## Roll-back Procedures

- Step: Revert Line Card Replacement, Command: replace board , Description: If replacing the line card does not resolve the issue or introduces new problems, revert to the original configuration by replacing the new line card with the original one, ensuring that any changes made are properly reversed.

- Step: Revert ASIC Reset, Command: reset asic to-default, Description: If resetting the ASIC does not resolve the issue, use the command to reset the ASIC to its default state, reverting any changes made during the attempted fix.

- Step: Disable Restored Power Supply, Command: power supply disable, Description: If restoring power supply redundancy does not resolve the issue, disable the power supply that was enabled during the resolution attempt to revert to the previous power configuration.